

Controlled solid-state crystallization with formamidinium acetate for fully vacuum-deposited perovskite solar cells

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Vacuum deposition is a promising route for the fabrication of perovskite solar cells (PSCs). However, vacuum-deposited PSCs have not yet matched the performance achieved by solution-processed devices. Here we improve the performance of fully vacuum-deposited PSCs by controlling the solid-state reaction pathway. We find that formamidinium acetate reacts directly with PbI_2 to form FAPbI_3 seed layers, substantially lowering the barrier to the solid-state conversion of precursors into the crystalline δ -phase and their subsequent transformation into the α -phase perovskite. Moreover, the excess acetate passivates grain boundary defects, thereby suppressing non-radiative recombination. As a result, the PSCs achieve a power conversion efficiency of 25.53% and an electroluminescence external quantum efficiency of 18.38%. Importantly, the solvent-free fabrication also enables excellent device stability, with the devices retaining over 95% of their initial power conversion efficiency after 1,000 h under the ISOS-L-1 protocol.

Perovskite solar cells (PSCs) have emerged as a promising next-generation photovoltaic technology and are steadily progressing towards commercialization^{1–3}. Vacuum deposition has emerged as a promising method for the scalable production of perovskite thin films and devices, owing to its precise process control, excellent film uniformity, use of non-toxic solvents and high equipment repeatability^{4–6}. Despite the dominance of vacuum deposition processes in the manufacturing of other commercial semiconductor thin films, research on vapour-phase processed hybrid perovskites remains heavily underrepresented. One of the core challenges restricting the performance of vacuum-processed perovskite photovoltaic devices lies in suboptimal film quality, stemming from difficulties in precursor diffusion, phase transition processes and limitations in controlling crystallization. In solution-based processing, intermediate phases (for example, $\text{DMSO} \cdot \text{PbI}_2$, $(\text{FA} \cdots \text{NMP})\text{PbI}_3$) formed through solvent–solute

coordination play a critical role by substantially reducing the energy required for the δ - to α -phase transition^{7–9}. However, such intermediate phases are absent during the solid-phase reaction from thermally evaporated solid-state precursors. Consequently, the reaction and crystallization process must be driven by thermal energy, while random molecular diffusion often leads to incomplete and inhomogeneous reactions^{10–12}, resulting in difficult precursor phase transformation and limited control over the crystallization process.

Developing tailored additives to regulate crystallization is an effective strategy for improving the quality of perovskite films. In solution-processed perovskites, acetate additives have been shown to reduce the formation energy and facilitate the $\delta \rightarrow \alpha$ phase transition. Developing additives that are specifically compatible with solid-phase perovskite formation might effectively lower the reaction energy barrier, fundamentally enhancing both the solid-state reaction kinetics

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and film crystallinity. However, studies on strategies for solid-state perovskite formation remain scarce. Representative examples include lead chloride (PbCl_2), where chloride ions serve as an additive to achieve 24.42% efficiency¹³. Another strategy employs KSCN , which enhances precursor conversion and passivates grain boundaries, enabling 18.34% efficiency in FAPbI_3 devices¹⁰.

Here we report highly efficient fully vacuum-deposited PSCs by introducing formamidinium acetate (FAAc) into the layer-by-layer thermally evaporated perovskite precursors. FAAc could directly react with lead iodide (PbI_2) to form the FAPbI_3 seed layer and notably reduce the energy barriers associated with solid-state reactions and phase transitions. This allows a lower reaction temperature, accelerates the crystallization process and promotes perovskite crystal growth. Furthermore, the excess acetate ions can passivate grain boundary defects in the perovskite thin films and suppress non-radiative recombination. The resulting fully vacuum-deposited PSCs achieved a power conversion efficiency (PCE) of 25.53%, while the electroluminescence (EL) external quantum efficiency (EQE) reached 18.38%. The high PCEs and EQEs of fully vacuum-deposited PSCs and perovskite light-emitting diodes (PeLEDs) outperform existing published results, to our knowledge. The PSCs also demonstrated excellent operational stability, exhibiting less than 5% efficiency loss relative to their initial PCE after 1,000 h of operation at room temperature. In addition, they showed excellent damp-heat stability (55 °C, 55% relative humidity (RH)), with efficiency loss remaining below 2% of the initial PCE after 3,000 h.

Controlling the solid-state reaction

Figure 1a illustrates the simple layer-by-layer thermal-evaporation process for fabricating perovskite precursor films with FAAc, which consists of ionic compounds of formamidinium cation ($[\text{HC}(\text{NH}_2)_2]^+$) and acetate anion (CH_3COO^-) (Fig. 1b). This low-temperature molten salt exhibits a melting point of -158°C (Supplementary Fig. 2). It is feasible to prepare FAAc thin films using thermal evaporation (Supplementary Figs. 3 and 4). The as-deposited perovskite precursor thin films were annealed on a hotplate at 175°C under ambient atmosphere (Fig. 1c and Supplementary Video 1). During the annealing process, the colour of films gradually changed from yellow to black, indicating the formation of the perovskite phase. Notably, the phase transition occurred noticeably faster in the FAAc-treated film compared with the control, with transition times of approximately 5 s and 15 s, respectively. To track component changes of perovskite thin films at different annealing temperatures, we conducted X-ray diffraction (XRD) tests from approximately 30°C to 170°C (20°C gap), monitoring characteristic peaks of PbI_2 , $\delta\text{-FAPbI}_3$ and $\alpha\text{-FAPbI}_3$. As shown in Fig. 1d and Supplementary Fig. 5, the introduction of FAAc appreciably reduced both the disappearance temperature of the PbI_2 peak and the onset temperature for the formation of the δ -phase, indicating that FAAc accelerated the solid-state diffusion reaction of the precursors. Furthermore, the complete phase transition from δ - to $\alpha\text{-FAPbI}_3$ was lowered from 130°C to 110°C , demonstrating that FAAc plays a key role in controlling the solid-state perovskite reaction by enhancing reaction activity and lowering the transition temperature. To further simulate and magnify this process, we prepared a powder mixture with the same precursor composition ratio as used in the thermally evaporated thin films (Fig. 1e,f and Supplementary Video 2). When a mixture of PbI_2 , formamidinium iodide (FAI), caesium iodide (CsI) and PbCl_2 was ground, the powder appeared yellow-brown in colour, consisting primarily of PbI_2 with only trace amounts of $\delta\text{-FAPbI}_3$. Even after heating, strong characteristic peaks of PbI_2 remained, with minimal $\alpha\text{-FAPbI}_3$ signals detected. By contrast, when 20 mol% FAAc was added to the mixture, distinct $\delta\text{-FAPbI}_3$ peaks appeared immediately after grinding. After heating for the same duration, the powder turned black, indicating the formation of large amounts of the perovskite phase. During the later stages of annealing, liquid droplets were observed deposited on the inner walls of the glass bottles. NMR analysis revealed this substance

to be volatilized FAAc (Supplementary Fig. 4). XRD analysis confirmed that the powder had almost completely converted into $\alpha\text{-FAPbI}_3$, while the grain size and uniformity of the perovskite were obviously improved (Supplementary Fig. 5). Density functional theory (DFT) calculations (Supplementary Information) further revealed that FAAc can lower the reaction energy barriers for the solid-state conversion from precursors to the δ -phase and subsequently to the α -phase, as shown in the corresponding energy diagram in Fig. 1g. These results support the conclusion that FAAc plays a key role in the solid-state perovskite formation process, thereby enhancing crystallization quality. We found that both MAAC and PbAc_2 exhibit similar effects, confirming the critical role of protic nature and acetate (Ac^-) in controlling the solid-state formation of perovskite (Supplementary Figs. 7–9).

Temperature-dependent Raman spectroscopy (Fig. 1h) revealed that FAAc and PbI_2 could spontaneously react at room temperature to form $\delta\text{-FAPbI}_3$, while heating above 120°C facilitated the formation of $\alpha\text{-FAPbI}_3$. This reaction pathway (equation (1)) was further confirmed by XRD measurements of FAAc/ PbI_2 bilayer structures (Fig. 1i and Supplementary Figs. 10–13).



The strong coordination interaction and hydrogen-bonding interaction between FAAc and PbI_2 clearly lowers the energy barrier for the reaction, making it much more favourable¹⁴ (Supplementary Figs. 14 and 15). As shown in Supplementary Fig. 16, physical mixing of FAAc and PbI_2 powders resulted in spontaneous interfacial reaction even at room temperature, which occurred much more readily than the direct reaction between FAI and PbI_2 . Given the volatility of PbAc_2 at elevated temperatures, substantial PbAc_2 residues are unlikely to remain in the perovskite films (Supplementary Fig. 17). As shown in Fig. 1j, the perovskite precursors undergo mutual diffusion and interfacial reactions under the thermal field. The interfacial reaction between FAAc and PbI_2 thin films leads to the formation of FAPbI_3 , which serves as nucleation sites for the subsequent solid-state crystallization of perovskite, thereby promoting high-quality crystal growth. This conclusion is further supported by the scaled-up powder-based experiments (Supplementary Fig. 18).

In situ crystallization processes

To investigate the phase evolution of perovskite thin films during reaction with FAAc, we performed in situ grazing-incidence wide-angle X-ray scattering (GIWAXS; Fig. 2a and Supplementary Fig. 19). Control samples exhibited negligible perovskite signals and continuous PbI_2 diffraction ($q = 0.92 \text{ \AA}^{-1}$) during the initial annealing stage, indicating a limited FAI– PbI_2 reaction. By contrast, after employing FAAc, the three-dimensional (3D) intermediate ($q \approx 10.5 \text{ nm}^{-1}$) formed instantaneously at the initial stage of annealing (175°C). Moreover, the time required for the solid-state precursor to fully transform into the α -phase was shortened from 19.2 s to 16.8 s. The accelerated phase transition observed in FAAc-perovskite thin films can be attributed to the dual function of FAAc, which lowered the phase transition energy and provided the FAPbI_3 seeds for the better film crystalline quality. This was further validated by in situ ultraviolet–visible (UV–vis) absorption measurements and the morphological evolution via surface and cross-sectional scanning electron microscopy (SEM) images (Supplementary Figs. 20 and 23). We used in situ photoluminescence (PL) to track the evolution of the perovskite intermediates during the early stages (0–2.4 s) of thermal annealing at 175°C (Fig. 2b). The control film showed a weak PL peak around 785 nm, indicating that only a trace amount of emissive 3D-like species was formed. For the FAAc sample, a strong PL intensity was initially observed at around 810 nm, indicating the formation of an FAPbI_3 seed layer on the surface via the direct reaction between FAAc and PbI_2 (ref. 15). The PL peak of the FAPbI_3 seed layer was blue-shifted relative to that of ideal phase-pure FAPbI_3 ($\sim 840 \text{ nm}$),

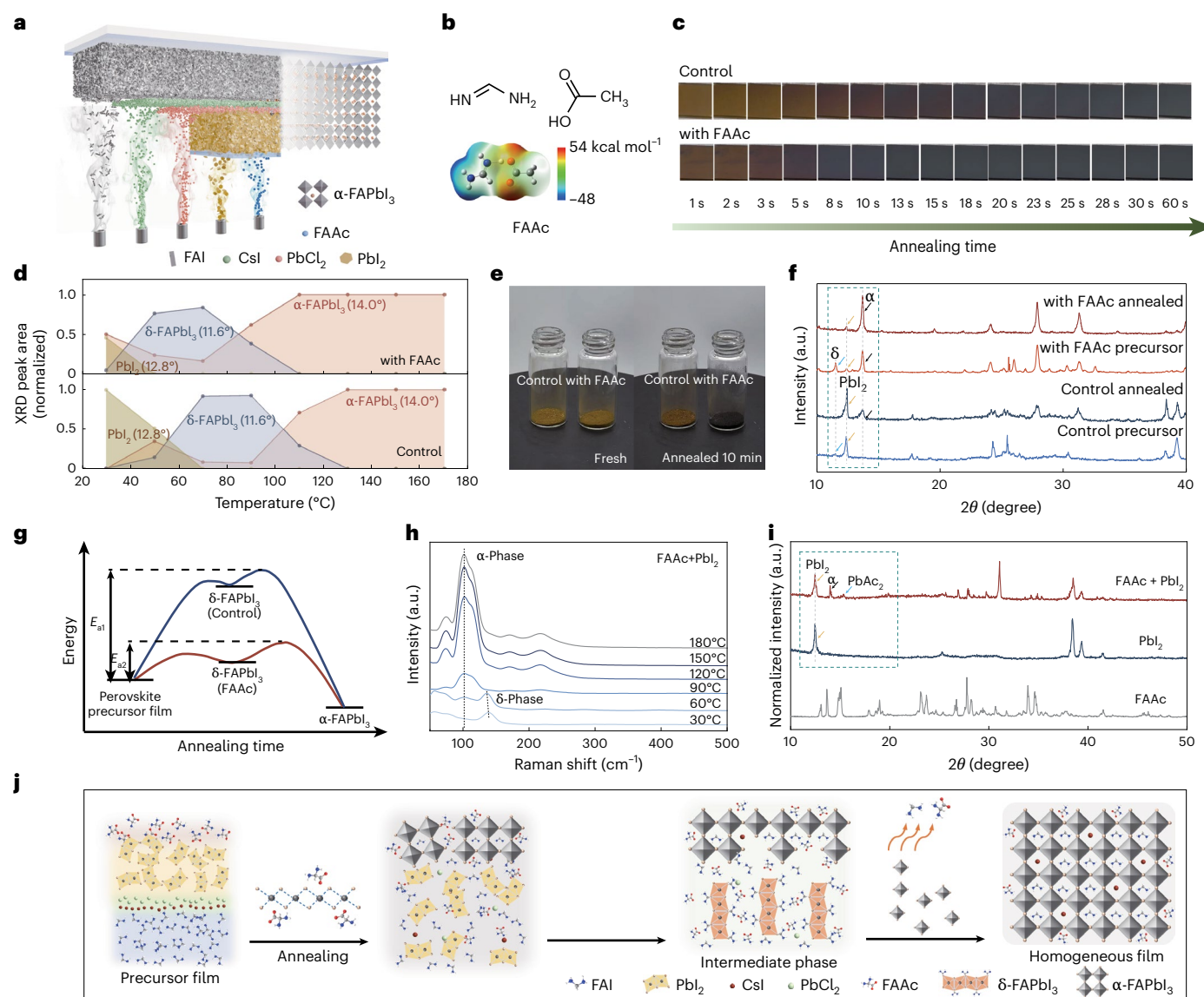


Fig. 1 | Solid-state reaction of the layer-by-layer evaporated perovskite thin films. **a**, Schematic illustration of the thermal evaporation of a perovskite thin film based on FAAC by a layer-by-layer deposition strategy. **b**, Chemical structures and electrostatic potential (ESP) maps of FAAC. **c**, Photographs illustrating the evolution of perovskite thin films during annealing (annealing temperature 175 °C). **d**, Integration of α -FAPbI₃, δ -FAPbI₃ and PbI₂ XRD peak areas at different reaction temperatures. **e**, **f**, Digital photographs (**e**) and XRD patterns (**f**) of PbI₂/FAI/CsI/FAAC mixed powders after mechanical milling and annealing at

175 °C. The blue, yellow and black arrows point to δ -FAPbI₃, PbI₂ and α -FAPbI₃, respectively. **g**, Scheme of energy variation as a function of reaction coordinate for the reaction FAI + PbI₂ → FAPbI₃ with or without FAAC. **h**, Temperature-dependent Raman spectra of mixed powders of PbI₂ and FAAC. **i**, XRD patterns of FAAC, PbI₂ and their mixed powders after annealing. The yellow, black and light-blue arrows point to PbI₂, α -FAPbI₃ and PbAc₂, respectively. **j**, Schematic diagram of the α -FAPbI₃ formation process with FAAC during the solid-state reaction and crystallization. The orange arrows represent the volatilization of excess FAAC.

which could be attributed to Cs⁺ and Cl⁻ incorporation and quantum confinement effects. As the reaction proceeded, a large amount of 3D intermediate phase formed, leading to a gradual blue shift of the PL peak to approximately 785 nm.

During annealing (175 °C), FAAC could diffuse downwards easily (Supplementary Figs. 21–23 and Supplementary Note 3). The FAAC–PbI₂ mixture could greatly accelerate interdiffusion with FAI (Supplementary Video 3). This effectively accelerated the solid-state reaction and improved the crystallinity and overall film quality of the thermally evaporated perovskite layer. FAAC-treated perovskite thin films exhibited appreciably enlarged grain sizes (Fig. 2c and Supplementary Fig. 24), whereas the control films showed smaller grains with obvious PbI₂ residues resulting from the inefficient solid-state reaction. Moreover, the pinholes at the buried interface were noticeably reduced, which facilitated more efficient charge extraction. This improvement

is primarily attributed to the enhanced compositional uniformity with involvement of FAAC.

As shown in Fig. 2d and Supplementary Fig. 25, the incidence angle-dependent grazing-incidence XRD (GIXRD) spectra of the (210) diffraction plane was collected at different incidence angles. For the control film, the diffraction peak position gradually shifted to higher 2 θ values with increasing incidence angle, indicating the presence of internal stress caused by vertical inhomogeneity in the film. By contrast, the FAAC-modified perovskite thin film showed negligible peak shift, suggesting that the film was free from internal stress and exhibited improved structural uniformity. As shown in Supplementary Fig. 26, for the control film, distinct Debye–Scherrer rings corresponding to PbI₂ were clearly observed in the 2D GIWAXS patterns collected at different incident angles, indicating a substantial amount of unreacted PbI₂ remaining in the bulk of the film. By contrast, for the FAAC-treated

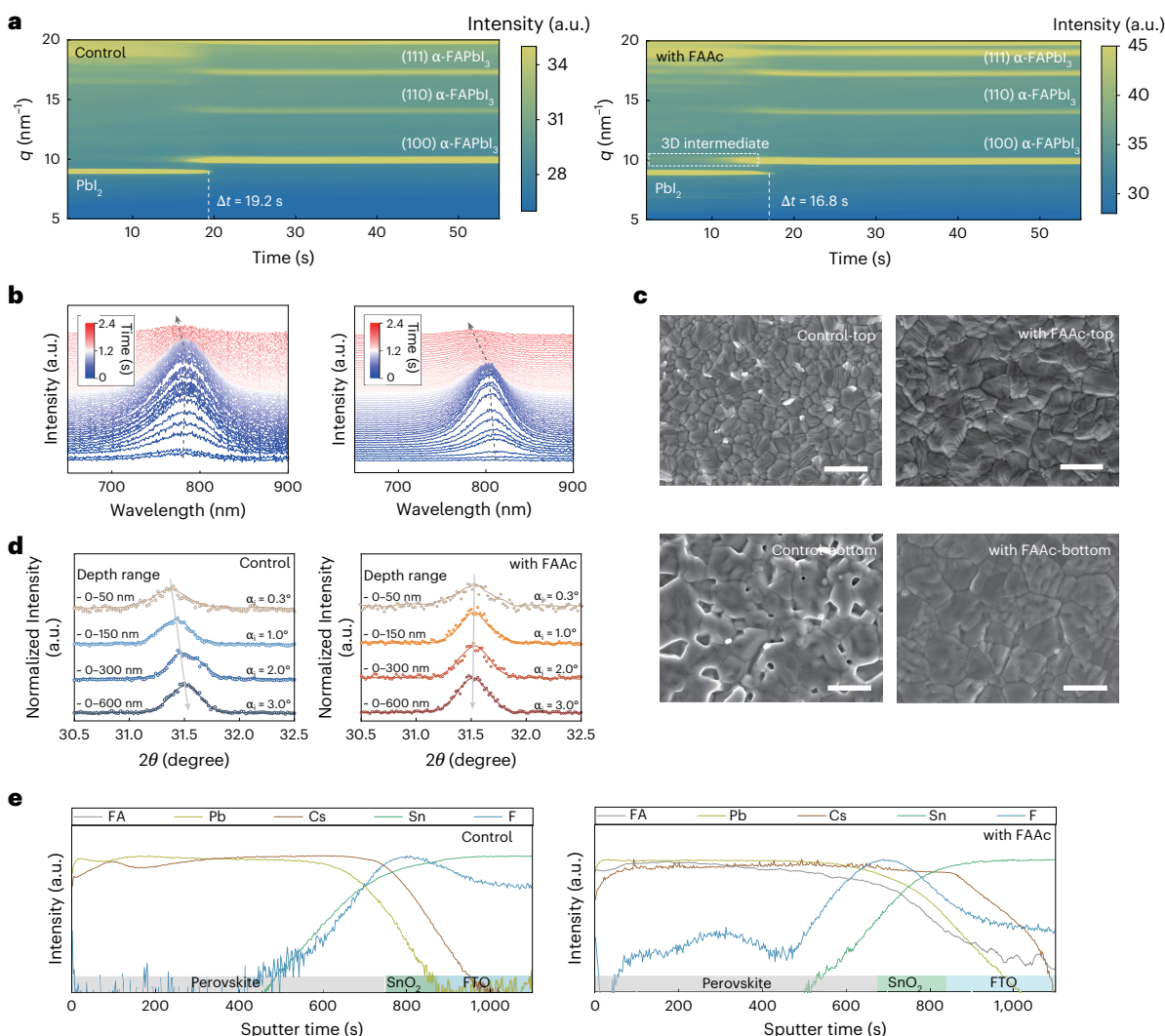


Fig. 2 | In situ crystallization processes and quality of perovskite thin films.

a, b. In situ GIWAXS (**a**) and in situ PL spectra (**b**) of perovskite precursor films during thermal annealing (annealing temperature 175 °C). The direction of the arrows represent the peak position change over time. **c.** Top-view SEM images of the top and bottom of the perovskite thin films. Scale bar, 1 μm . **d.** GIXRD

patterns of perovskite thin films at different probing depths, with Gaussian fitting of the diffraction peaks. The direction of the arrows represents the peak position variation of the perovskite thin films from top to bottom. **e.** Depth profiles of ion distribution in perovskite thin films, obtained by time-of-flight secondary ion mass spectrometry (ToF-SIMS).

film, no obvious PbI_2 signals were detected at any incidence angle. Moreover, the diffraction ring intensity of the perovskite was greatly enhanced, confirming the improved crystallinity and more complete conversion of precursors. The enhancement of the vertical distribution of perovskite components was further demonstrated by the ToF-SIMS pattern of the perovskite thin films (Fig. 2e). In the control sample, the signal intensities of Pb and Cs exhibit notable fluctuations, indicating uneven element distribution. By contrast, the target sample exhibits excellent compositional uniformity¹⁶. The remarkably flat and stable elemental signals indicate that the addition of FAc enhances the crystalline quality of perovskite thin films, achieves more uniform elemental distribution and reduces bulk defects.

Bulk passivation

The perovskite surface is highly prone to the formation of uncoordinated Pb^{2+} ions, leading to the formation of abundant interfacial defects. The strong coordination interaction between Ac^- and Pb^{2+} offers an effective strategy for interfacial defect passivation. Theoretical calculations (Fig. 3a) revealed that Ac^- could strongly adsorb onto the exposed Pb^{2+} sites, elongating the Pb–I bond length from 3.079 Å to 4.13 Å. This interaction effectively reduced the density of trap states within the

bandgap (Fig. 3b), thereby suppressing non-radiative recombination and mitigating efficiency loss¹⁷. Compared with the control sample, the Pb 4f and I 3d core levels in the FAc-treated perovskite exhibited a shift towards lower binding energies (Fig. 3c and Supplementary Fig. 27), indicating a stronger chemical interaction¹⁸. In addition, the red-shifted O–H bond in the Fourier-transform infrared spectroscopy confirmed the strong coordination between Ac^- and the perovskite lattice (Supplementary Fig. 28). This interaction effectively reduced the trap density from 7.7×10^{16} to $6.0 \times 10^{16} \text{ cm}^{-3}$ in the perovskite thin films, as indicated by the lower trap-filled limit voltage (V_{TFL}) observed in single-carrier devices¹⁹ (Supplementary Fig. 29). To further evaluate the defect characteristics, we conducted temperature-dependent impedance spectroscopy to extract the trap energy levels^{20,21}. Based on the Arrhenius fitting (Fig. 3d and Supplementary Fig. 30), the defect activation energies (E_a) for the control and FAc devices were calculated to be 11.66 meV and 7.81 meV, respectively, indicating a shallower trap level in the FAc device.

The introduction of FAc also contributed to improved uniformity of the perovskite thin films. First, we measured the PL intensity from both the top and bottom surfaces of the perovskite thin films. As shown in Supplementary Fig. 31, the FAc perovskite exhibited nearly identical

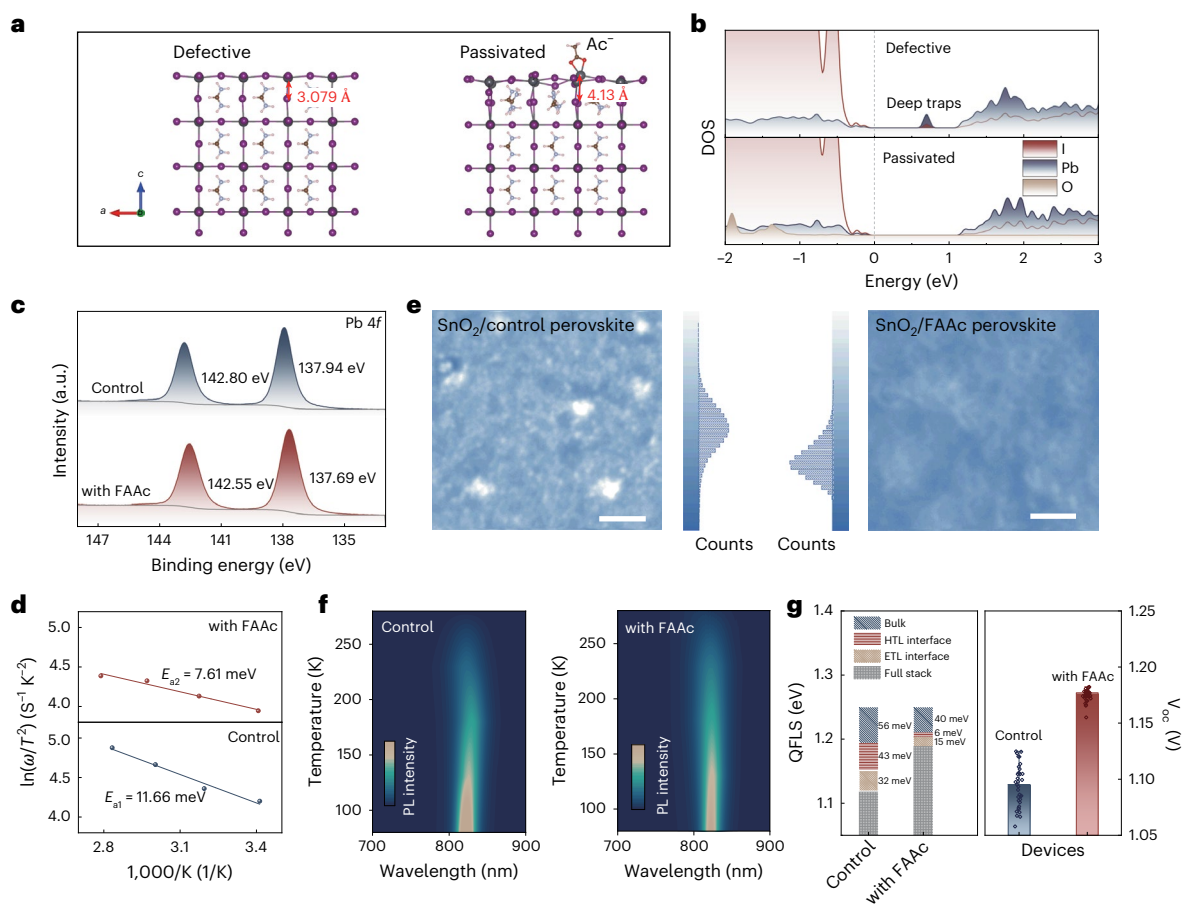


Fig. 3 | Defect passivation of perovskite thin films. **a**, A $3 \times 3 \times 1$ slice of the optimized structures of the control and Ac^- -passivated perovskite surfaces; FA^+ ions located at surfaces or interfaces exhibit rotational motion due to asymmetric chemical environments or defects. Legend for atom labelling: grey, Pb; red, O; brown, C; purple, I; blue, N; pink, H. **b**, The density of states (DOS) distribution for the control and Ac^- -modified perovskite with iodine vacancy. **c**, XPS curves of perovskite thin films, where the peaks are fitted with Gaussian functions. **d**, Arrhenius plot of the characteristic frequencies at different temperatures extracted from the derivative specific capacitance spectra.

The solid lines represent the fitting results with Arrhenius function. **e**, PL mapping of perovskite thin films, with intensity distribution plots on the sides. Scale bar, 10 μm . **f**, Pseudo-colour maps illustrating temperature-dependent PL intensity in the range of 80–300 K. **g**, Statistics for QFLS in half-devices and V_{oc} statistics for devices. Left two columns: QFLS statistics calculated for perovskite thin films with and without charge transport layers (CTLs); right two columns: V_{oc} statistics for devices ($n = 35$ samples). The centre line within each box represents the median value.

PL peak positions and full width at half maxima on both surfaces, indicating enhanced vertical compositional uniformity compared with the control. We further assessed in-plane uniformity using PL mapping (Supplementary Fig. 32). The intensity and distribution uniformity of both PL and time-resolved PL mapping were notably enhanced in the FFAc perovskite thin films on glass substrates compared with the control samples. These results suggested that FFAc not only enhanced film uniformity but also effectively passivated defects and suppressed non-radiative recombination, which aligned with the above conclusions. In addition, we examined the PL intensity distribution of perovskite thin films deposited on tin dioxide (SnO_2) substrates. As shown in Fig. 3e, the intensity of the PL mapping was reduced after incorporation of FFAc, which indicated the more efficient charge extraction. This contributes to enhancing the photovoltaic performance of the devices. This was further confirmed by the temperature-dependent PL measurements (Fig. 3f and Supplementary Fig. 33). By fitting the pseudo-colour maps in Fig. 3f, the exciton binding energies of the control and FFAc perovskite were determined to be 105.37 and 86.28 meV, respectively. The lower exciton binding energy in the FFAc perovskite facilitated exciton dissociation and accelerated charge extraction²².

To assess the impact of trap-induced non-radiative recombination on voltage loss, we calculated the quasi-Fermi level splitting (QFLS) of

the perovskite thin films using PL quantum yield (PLQY) data^{16,23} (Fig. 3g and Supplementary Fig. 34). Under illumination, the QFLS of perovskite materials determines the maximum possible open-circuit voltage (V_{oc}) achievable under ideal conditions (that is, without interfacial losses), which could be calculated through the measurement of PLQY. The higher QFLS value observed in FFAc-treated samples directly demonstrated reduced bulk non-radiative recombination within the material itself. This improvement was attributed to enhanced crystallinity and the effective passivation of defects by Ac^- . Contact loss represents the difference between the QFLS of the perovskite bulk and the actual V_{oc} ultimately measured across the cell terminals. The contact losses at the electron transport layer (ETL) and hole transport layer (HTL) interfaces decreased from 32 meV and 73 meV to 15 meV and 6 meV, respectively. This indicated that non-radiative recombination loss was suppressed after FFAc employment.

Device characteristics

To enable the fabrication of fully vacuum-processed PSCs, we developed a magnetron sputtering technique for depositing SnO_2 (M- SnO_2) as the ETL and thermally evaporated Spiro-TTB:MoO₃ as the HTL (Supplementary Fig. 35). As shown in Fig. 4a and Supplementary Figs. 36–38, M- SnO_2 exhibited comparable optical transmittance and

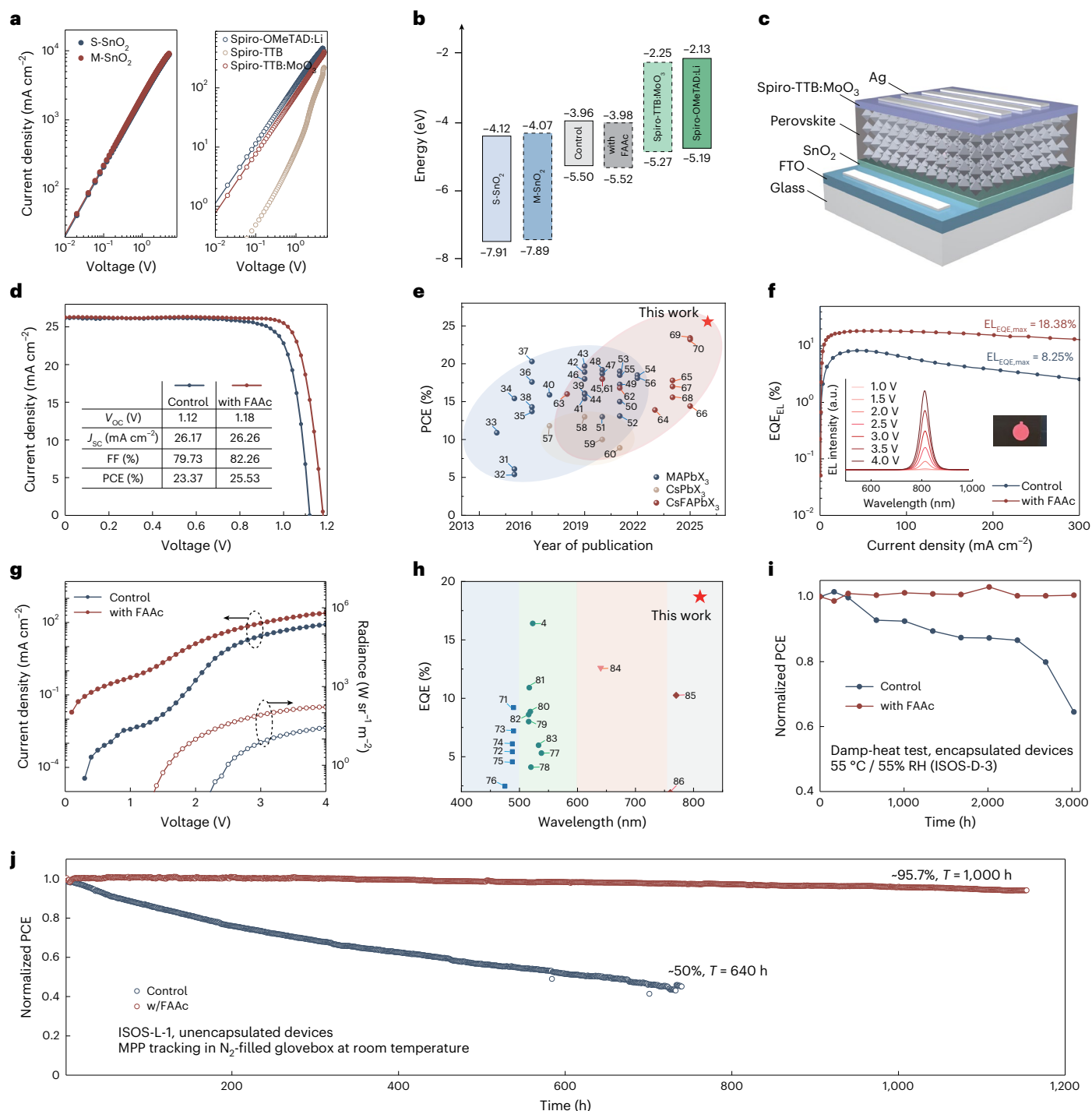


Fig. 4 | Device structure and optoelectronic performance. **a**, Current density–voltage (J – V) curves for single-carrier devices. The curves are guides to the eye. **b**, Energy-level alignment of PSCs. **c**, Schematic illustration of the device structure. **d**, J – V curves of the champion PSCs fabricated by fully vacuum deposition. **e**, Evolution of PCE for fully vacuum-deposited PSCs reported over the years^{12,31–70} (detailed photovoltaic parameters are provided in Supplementary Tables 1–3). **f**, EL EQE–current density curves of the PSC when working as an LED, the inset shows the EL spectra at different voltages and its emission image. **g**, Current density–voltage and radiance–voltage characteristics of the PSC

when working as an LED. **h**, Summary of recently reported EQE for thermally evaporated PeLEDs^{4,71–86} (detailed photovoltaic parameters are provided in Supplementary Tables 4–7). **i**, Long-term stability measurements of the encapsulated PSCs under damp-heat storage conditions (55 °C/55% RH). Their initial PCEs for the control and target were 22.29% and 24.83%, respectively. **j**, The MPP tracking of unencapsulated PSCs under continuous 1-sun illumination when stored in an N₂-filled glovebox (25 °C). The light source is a white-light LED with its intensity calibrated to 1-sun illumination (Supplementary Fig. S1). Their initial PCEs for the control and target PSCs were 22.35% and 25.14%, respectively.

higher electron mobility than conventional solution-processed films (S-SnO₂). Moreover, the conduction band minimum of M-SnO₂ (–4.07 eV) aligns more closely with that of the perovskite layer (–3.98 eV), thereby reducing the interfacial energy barrier during electron extraction

(Fig. 4b and Supplementary Figs. 39 and 49). These advantages allow the M-SnO₂ to facilitate efficient charge collection in PSCs. For the HTLs, we employed thermally evaporated Spiro-TTB doped with molybdenum trioxide (MoO₃) as a replacement for the conventional Li-TFSI-doped

Spiro-OMeTAD. As shown in Fig. 4a, the hole mobility of thermally evaporated pristine Spiro-TTB is lower than that of spin-coated Spiro-OMeTAD:Li. To enhance the conductivity of Spiro-TTB, we introduced MoO₃, a widely used p-type dopant in organic HTLs²⁴. MoO₃ doping substantially increases the conductivity of Spiro-TTB, achieving level comparable to that of Spiro-OMeTAD:Li. In addition, the valence band maximum of Spiro-TTB:MoO₃ (−5.27 eV) is closer to that of the perovskite layer (−5.52 eV) (Supplementary Figs. 43 and 44), ensuring efficient hole transport. Thermal evaporation Spiro-TTB:MoO₃ provides excellent coverage while preserving the initial morphology of the perovskite thin film (Supplementary Figs. 41 and 42), owing to its thickness of only 30 nm. By contrast, the spin-coated Spiro-OMeTAD:Li film has a thickness of approximately 200 nm (Supplementary Fig. 40). Both HTLs exhibit complete coverage, effectively preventing leakage in the devices. Furthermore, the thermally evaporated Spiro-TTB:MoO₃ exhibited excellent thermal stability (decomposition temperature exceeding 530 °C; Supplementary Fig. 45), and hydrophobicity (Supplementary Fig. 46), both of which contributed to enhanced device stability. The favourable optoelectronic properties of both M-SnO₂ and evaporated Spiro-TTB:MoO₃ established a solid foundation for the development of high-performance, fully vacuum-deposited PSCs.

As shown in Fig. 4c, the optimized device structure is FTO/M-SnO₂ (20 nm)/perovskite active layer (700 nm)/Spiro-TTB:MoO₃ (25 nm)/MoO₃ (5 nm)/Ag (100 nm). The control and target perovskite film shared similar band gaps (Supplementary Figs. 47 and 48). The preferred annealing humidity is around 30% RH. Although FAAC facilitates intermolecular diffusion, a certain level of humidity remains essential (Supplementary Note 5 and Supplementary Figs. 49 and 50). We investigated the impact of FAAC on the performance of the resulting PSCs. The current density–voltage (*J*–*V*) characteristics were measured under 1 sun. As shown in Fig. 4d and Supplementary Fig. 52, the *V*_{oc} of the champion FAAC-PSC was vastly improved from 1.12 to 1.18 V, which could be attributed to the elimination of buried interfacial voids, enhanced precursor diffusion and reaction kinetics, and reduced defect intensity. The fill factor (FF) was also improved to 82.26%, indicating the improved crystallinity of the perovskite thin film. The high short-circuit current density (*J*_{sc}) was guaranteed by the efficient charge mobility and light absorption (Supplementary Figs. 53 and 54), which in good agreement with the integrated *J*_{sc} in the EQE spectrum (Supplementary Fig. 55). The champion FAAC-based PSC achieved a PCE of 25.53%, outperforming previously reported efficiencies for fully vacuum-deposited PSCs, to our knowledge (Fig. 4e and Supplementary Tables 1–3). We also achieved a third-party laboratory-certified PCE of 24.93% (*J*_{sc} of 26.08 mA·cm^{−2}, a *V*_{oc} of 1.16 V and a FF of 82.65%) (Supplementary Fig. 56). Compared with solution-based methods, the fully vacuum-based process enabled much better reproducibility due to its high controllability and uniformity (Supplementary Note 6 and Supplementary Fig. 57–59), which was critical for scalable and reliable device production. The 1.00-cm² device achieved a PCE of 24.00% (Supplementary Fig. 60), with only a slight efficiency loss compared with the small-area device. We further found that FAAC is equally effective in the p–i–n architecture (Supplementary Note 7 and Supplementary Fig. 61). Notably, the seed-layer strategy can also operate at the buried interface, demonstrating the generality and versatility of this approach for regulating solid-state perovskite crystallization (Supplementary Note 8 and Supplementary Fig. 62).

Moreover, the PSC exhibited stable and bright near-infrared EL with an emission peak at 813 nm (Fig. 4f). FAAC perovskite thin films exhibit high crystallinity, with their EL peaks remaining stable even under high electric fields. By contrast, control devices show EL shifts with increasing bias voltage, probably due to halide migration (Supplementary Fig. 63). The device showed a low turn-on voltage of −1.3 V (Fig. 4g), demonstrating efficient charge injection and transport within the device. The high-quality perovskite films ensure a high radiative recombination efficiency (η_{rad} ; Supplementary Fig. 64), enabling the FAAC-based device to achieve an EQE of 18.38% (Fig. 4f).

However, the control device exhibited larger turn-on voltage (−2.2 V), inferior EQE (8.26%) and irradiance (43.71 W sr^{−1} m^{−2}). This superior EL performance highlighted the excellent optoelectronic quality of the thermally evaporated perovskite thin films, which was attributed to improved crystallinity and suppressed non-radiative recombination. Figure 4h and Supplementary Tables 4–7 summarize the performance of recently reported thermally evaporated perovskite light-emitting diodes (LEDs) (ranging from visible to near-infrared emission). The EL efficiency obtained with the FAAC strategy represents a notable improvement over the state of the art. These results demonstrated the huge potential of this strategy in fabricating high-performance perovskite optoelectronic devices.

Finally, to evaluate the stability of the perovskite thin films, we monitored the evolution of the PL spectra under continuous laser (445 nm) irradiation. As shown in Supplementary Fig. 65, the PL intensity of the FAAC perovskite thin films with increased crystallinity exhibited excellent stability, maintaining 80% of the initial intensity after 800 min. The thermogravimetric response of the perovskite powder also verified the positive effect of FAAC on the stability of perovskite crystals (Supplementary Fig. 66). Subsequently, we performed a series of stability studies on the corresponding devices. The encapsulated cells were tested for damp heat (55 °C and 55% RH, ISOS-D-3 protocol)²⁵. As shown in Fig. 4i, the FAAC device showed almost constant PCE after 3,000 h, while the control device retained only 60% of its initial PCE. In addition, to confirm the operational stability of the fully vacuum-deposited devices, the unencapsulated samples were exposed to continuous aging tests in a glove box, following as ISOS-L-1 protocol. As shown in Fig. 4h, the FAAC device maintained 95.7% of its initial PCE after 1,000 h of continuous operation, while the control device exhibited a more pronounced degradation, retaining only 50% of its initial performance after approximately 640 h. As shown in Supplementary Table 8, the performance of fully vacuum-deposited PSCs is approaching the highest level achieved by solution-processed devices. In addition to the improved quality of the perovskite thin film, the enhanced device stability was also attributed to the excellent intrinsic stability of the thermally evaporated Spiro-TTB:MoO₃ HTL and the entirely solvent-free, fully vacuum-based fabrication process. This work not only demonstrated the effectiveness of FAAC-manipulated solid-state crystallization but also showcased a scalable route for fabricating efficient and stable PSCs, paving the way for future commercial applications of vacuum-processed perovskite optoelectronic devices.

Conclusions

In summary, we realized the efficient and luminescent PSCs via a fully vacuum-deposition technique by introducing FAAC to regulate the solid-state reaction and crystallization, and engineering vacuum-compatible charge transport layers. FAAC effectively reduced the energy barrier for phase transition and enabled the formation of FAPbI₃ seed layers, which notably improved perovskite crystallinity and film quality. Furthermore, the passivation effect of excess Ac[−] ions suppressed non-radiative recombination, contributing to enhanced optoelectronic properties. We also developed efficient and stable magnetron-sputtered SnO₂ and thermally evaporated MoO₃-doped Spiro-TTB. This enabled the fabrication of fully vacuum-deposited n–i–p PSCs, achieving a champion PCE of 25.53%. Notably, the same architecture could also work as a highly efficient NIR perovskite LED with an EQE_{EL} of 18.38%, highlighting the versatility of our strategy. The enhanced perovskite thin film quality and fully vacuum-deposited fabrication process contributed to outstanding device stability, which maintained 95.7% of its initial PCE after 1,000 h of continuous operation. This work not only demonstrated the effectiveness of FAAC in controlling solid-state crystallization, but also showcased a scalable route for fabricating efficient and stable PSCs, paving the way for future commercial applications of vacuum-processed perovskite optoelectronic devices.

Methods

Materials

FAI was purchased from GreatCell Solar Materials. FAAC, PbI_2 , CsI, PbCl_2 and MoO_3 were purchased from TCI. Spiro-TTB was purchased from Bide Pharmatech. SnO_2 target (99.99% purity) and high-purity silver were purchased from ZhongNuo Advanced Material Technology. All materials were used as received without further purification. The 1.1-mm glass substrates patterned with F-doped tin oxide (FTO) ($7\ \Omega$ per square) were received from SJNYO.

Magnetron-sputtered SnO_2 thin films

SnO_2 thin films were deposited on FTO substrates by magnetron sputtering at room temperature. The cleaning method for FTO substrates involves sequential ultrasonic cleaning in detergent, deionized water and ethanol, with each solution cycle lasting 20 min, followed by nitrogen drying. The substrates were placed into the sputter chamber by first evacuating the pressure to 5.0×10^{-4} Pa. Before deposition, the SnO_2 target was presputtered for 10 min in a pure argon (Ar) atmosphere to remove contaminants from the target surface. The SnO_2 targets were then subjected to a constant radio-frequency power of 80 W for a sputtering time of 5 min. During sputtering, the argon flow was maintained at 10 sccm and the working pressure at 2 Pa. The substrate stage was rotated continuously at 15 revolutions per minute to maintain uniformity across the film.

Evaporation of perovskite thin films

The perovskite thin film was deposited via sequential layer-by-layer thermal evaporation (Shenyang Qihui Vacuum Technology). The pathway is the sequential evaporation of FAI/CsI/ PbCl_2 / PbI_2 on ITO/ SnO_2 substrates. The FAI powder was placed on a separate quartz crucible, and the evaporation rate was monitored by a QCM next to the substrate at $1.0\ \text{\AA}\ \text{s}^{-1}$ (Supplementary Fig. 1). The final thickness of FAI was 370 nm, and the initial deposition temperature was $-120\ ^\circ\text{C}$; 15 nm CsI, 20 nm PbCl_2 , and 400 nm PbI_2 were sequentially evaporated using rates of 0.2, 0.5 and $6.0\ \text{\AA}\ \text{s}^{-1}$, respectively. A 10 nm layer of FAAC was subsequently thermally evaporated onto the surface of the unannealed perovskite precursor film at a rate of $0.1\ \text{\AA}\ \text{s}^{-1}$. The deposited precursor films were annealed at $175\ ^\circ\text{C}$ for 10 min under atmospheric conditions (RH ~ 20 –30%) to obtain perovskite thin films. The calculation converting the thickness of the thermally evaporated precursor film into a molar ratio yields the final perovskite thin film composition as $\text{Cs}_{0.05}\text{FA}_{0.95}\text{PbI}_3$.

Device fabrication

For devices with the structure of FTO/ SnO_2 /perovskite/Spiro-TTB: MoO_3 / MoO_3 /Ag. Thermal evaporation was performed under high-vacuum conditions ($<5 \times 10^{-4}$ Pa). After the perovskite thin films were cooled to room temperature, they were transferred to another evaporation system to prepare the HTLs and the electrode layer. For the preparation of the HTL, 25 nm of Spiro-TTB and MoO_3 were co-evaporated at rates of 0.20 and $0.01\ \text{\AA}\ \text{s}^{-1}$, and then a 5-nm layer of pure MoO_3 (evaporation rate $0.10\ \text{\AA}\ \text{s}^{-1}$) continued to be evaporated on top; 100 nm silver was deposited via thermal evaporation as the back electrode. Finally, after flipping the device, 100 nm MgF_x was deposited onto the glass side of the substrate at a rate of $0.50\ \text{\AA}\ \text{s}^{-1}$ to serve as an anti-reflective coating that reduces light reflection loss.

Device measurement

The current density–voltage (J – V) characteristics were measured using a Keithley 2400 source meter under illumination from a solar simulator (Newport, Class A) with a light intensity of $100\ \text{mW}\ \text{cm}^{-2}$ (checked with a calibrated reference solar cell from Newport). The J – V curves were measured in a nitrogen atmosphere with a scanning rate of $100\ \text{mV}\ \text{s}^{-1}$, with a voltage step of 0.02 V and delay time of 1 ms. The active area was determined by the aperture shade mask ($0.066\ \text{cm}^2$) placed in front of the solar cell. A spectral mismatch factor of 1.0 was used for all J – V

measurements. The EQE was measured in ambient air using a solar-cell spectral response measurement system (QE-R, EnliTech).

Stability characterization

The operational stability tests (maximum power point (MPP) tracking) were performed using a multichannel thin-film photovoltaic attenuation test system (Suzhou D&R Instrument, PVL 6001M-16A) under an LED lamp adjusted to AM 1.5 G illumination ($100\ \text{mW}\ \text{cm}^{-2}$).

Damp heat stability tests were conducted by putting devices at $55\ ^\circ\text{C}/55\%\ \text{RH}$ in the environment test chamber. Devices for damp-heat stability testing require encapsulation. A uniform layer of edge sealant (butyl rubber) is applied to the surface of the PSCs. A glass cover is placed on top and cold-pressed for 12 h to prevent moisture ingress. Subsequently, the cell is re-encapsulated using an epoxy-acrylate UV-curable adhesive around the perimeter to ensure a tight seal. The solar cell electrodes are connected using silver tape, with this process completed inside a nitrogen glove box.

Crystallographic characterizations

The XRD spectra were measured on a Bruker D8 Advance X-ray diffractometer at room temperature using $\text{Cu-K}\alpha$ radiation at 40 kV and 40 mA. The wavelength of X-rays was $1.540593\ \text{\AA}$. GIWAXS was performed by employing a beam energy of keV and a PILATUS detector at the BL02U2 beamline of Shanghai Synchrotron Radiation Facility, Shanghai, China. GIWAXS patterns were acquired using an incident angle of 0.3 – 1.1° , and the sample was positioned approximately 241 mm from the detector (Pilatus).

Other characterizations

NMR spectra were recorded by Bruker Avance III HD500. X-ray photo-electron spectroscopy (XPS) measurements were conducted by a ThermoFisher Nexsa spectrometer. The surface and cross-sectional morphologies of the films and devices were characterized by field-emission SEM (SU8010, HITACHI) at an accelerating voltage of 5.0 kV and a beam current of 9,800 nA. The atomic force microscopy and conductive atomic force microscopy measurements were performed using a Bruker Dimension Icon atomic force microscope in an Ar-filled glovebox.

Space-charge-limited current measurements were performed on fully thermally evaporated electron-only and hole-only devices with the configurations FTO/2PACz/perovskite/Spiro-TTB: MoO_3 / MoO_3 /Ag, respectively. The J – V characteristics of space-charge-limited current measurements were recorded from 0 to 5 V with a 0.02 V step size under dark conditions using a Keithley 2400 source/meter unit.

Steady-state PL and time-resolved PL were measured using an Edinburgh FLS980 system. The light was illuminated from the top surface of the perovskite thin film. The fluorescence spectrometer used had an excitation wavelength of 480 nm. The repetition rate was 100 kHz. The laser fluence was $19.2\ \text{mJ}\ \text{cm}^{-2}$, and the power density was $0.0019\ \text{W}\ \text{cm}^{-2}$.

The temperature-dependent PL measurement were conducted as a continuous process throughout the entire measurement range from 298 K to 80 K. The temperature was lowered at a rate of $20\ \text{K}\ \text{min}^{-1}$ with holding points (maintained for approximately 10 min) set for testing, with each sample taking about 1 h. This time control was implemented to accurately reflect temperature effects rather than material aging effects. The UV–vis spectrometer (PerkinElmer model Lambda 2S) system was used to record UV–vis absorptions of films. The measurements were conducted over a wavelength range of ~ 300 – $900\ \text{nm}$ with a step size of 1 nm. The PL and time-resolved PL mapping were performed using a FILM 300 system (TIME TECH SPECTRA).

Trap density

The defect state values of the devices are calculated using the formula $V_{\text{TFL}} = eNL^2/(2\varepsilon_0\varepsilon)$, where V_{TFL} denotes the trap filling voltage, L

represents the perovskite thickness, N is the defect state density, ϵ is the relative permittivity of perovskite (46.9) and e is the elementary charge.

Exciton binding energy

The excitation energy E of the perovskite was further extracted through its temperature-dependent PL spectrum. The integrated PL intensity exhibited a monotonically increasing trend with decreasing temperature. Fitting was performed using

$$I(T) = I_0 / [1 + A \times \exp(-E_a / K_B \times T)], \quad (2)$$

where $I(T)$ is the integrated PL intensity at temperature T , I_0 is the PL intensity at 0 K, E_a is the activation energy and K_B is the Boltzmann constant.

Calculation of QFLS

QFLS can be calculated from PLQY by

$$\text{QFLS} = \text{QFLS}_{\text{rad}} + K_B \ln(\text{PLQY}) = K_B \ln \left(\frac{J_G}{J_{0,\text{rad}}} \times \text{PLQY} \right), \quad (3)$$

where QFLS_{rad} is the radiation limit of semiconductor materials, which sets the maximum achievable splitting of the quasi-Fermi level without considering non-radiative radiation recombination; K_B is the Boltzmann constant; J_G is the photogenerated current density; and $J_{0,\text{rad}}$ is the dark-state radiative recombination saturation current density. According to the detailed balance theory, $J_{0,\text{rad}}$ can be calculated by the following equations:

$$J_{0,\text{rad}} = q \int_0^\infty \text{EQE}_{\text{PV}}(E) \phi_{\text{BB}}(E) dE \quad (4)$$

$$\phi_{\text{BB}}(E) = \frac{2\pi e^2}{h^3 c^2 \exp\left(\frac{E}{k_B T}\right) - 1}, \quad (5)$$

where q is the elementary charge, EQE_{PV} is the photovoltaic EQE, ϕ_{BB} is the black-body radiative spectrum, E is the photo energy, h is the Planck constant and c is the light speed in vacuum.

DFT calculations

DFT calculations for adsorption and activation energies were performed using the Vienna Ab initio Simulation Package²⁶. To accurately describe the electronic exchange and correlation effects of the system, the widely used Perdew–Burke–Ernzerhof functional within the generalized gradient approximation was employed. The projector-augmented wave method was adopted to enhance computational precision and reliability²⁷. The plane-wave energy cut-off was set to 520 eV, with an energy convergence criterion of 1×10^{-5} eV and a force convergence threshold of 0.02 eV Å⁻¹. In the simulations, the α -FAPbI₃ perovskite phase was modelled using a cubic unit cell with lattice parameters $a = b = c = 6.362$ Å (ref. 28), while the δ -FAPbI₃ phase was constructed based on the $P6_3mc$ crystal structure.

The reaction energy $\Delta E_{\text{(reaction)}}$ of the chemical process was defined as

$$\Delta E_{\text{(reaction)}} = E_{\text{(products)}} - E_{\text{(reactants)}}. \quad (6)$$

Specifically

$$\Delta E_{\text{(reaction)}} = E_{\text{(FAPbI}_3\text{)}} - E_{\text{(PbI}_2\text{)}} = -18.89 \text{ eV} = -435.58 \text{ kcal mol}^{-1}.$$

Upon introducing 10% Ac⁻ into δ -FAPbI₃, the formation energy $\Delta E_{\text{(reaction)}}$ was calculated as

$$\Delta E_{\text{(formation)}} = E_{\text{(total)}} - E_{\text{(molecular reference)}}, \quad (7)$$

where the pristine δ -FAPbI₃ showed

$$\Delta E(\delta\text{-FAPbI}_3) = -1,282.96 \text{ eV}$$

and the AC⁻-modified δ -FAPbI₃

$$\Delta E(\delta\text{-FAPbI}_3 - \text{Ac}) = -1294.38 \text{ eV}.$$

Compared with the pristine δ -FAPbI₃, the incorporation of 10% acetate lowers the formation energy by

$$\Delta E = -11.42 \text{ eV} = -263.19 \text{ kcal mol}^{-1}.$$

For the adsorption models, a $3 \times 3 \times 3$ supercell of α -FAPbI₃ was employed, with a 15 Å vacuum layer introduced along the a -direction to avoid spurious interactions. Geometry optimizations were performed using a $1 \times 1 \times 1$ Γ -centred k -point mesh for Brillouin zone sampling, while electronic structure calculations used a denser $1 \times 3 \times 3$ k -point grid. The DFT-D3 dispersion correction was included to account for van der Waals interactions. In addition, for the electrostatic potential (ϕ) analysis of FA⁺ and AC⁻, calculations were performed at the B3LYP/6-311G level of theory using the Gaussian16 software package²⁹, and the resulting electrostatic surface potentials were analysed with Multiwfn to extract the maximum (ϕ_{max}) and minimum (ϕ_{min}) values³⁰.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The data that support the findings of this study are available within the Article and its Supplementary Information. Source data are provided with this paper.

References

- Yuan, L. et al. Perovskite solar cells and light emitting diodes: materials chemistry, device physics and relationship. *Chem. Rev.* **125**, 5057–5162 (2025).
- Best research-cell efficiency chart. NLR <https://www.nlr.gov/pv/cell-efficiency> (2026).
- Chen, P. et al. Multifunctional ytterbium oxide buffer for perovskite solar cells. *Nature* **625**, 516–522 (2024).
- Li, J. et al. Efficient all-thermally evaporated perovskite light-emitting diodes for active-matrix displays. *Nat. Photon.* **17**, 435–441 (2023).
- Abzieher, T. et al. Vapor phase deposition of perovskite photovoltaics: short track to commercialization?. *Energy Environ. Sci.* **17**, 1645–1663 (2024).
- Kosasih, F. U., Erdenebileg, E., Mathews, N., Mhaisalkar, S. G. & Bruno, A. Thermal evaporation and hybrid deposition of perovskite solar cells and mini-modules. *Joule* **6**, 2692–2734 (2022).
- Zou, Y. et al. A crystal capping layer for formation of black-phase FAPbI₃ perovskite in humid air. *Science* **385**, 161–167 (2024).
- Yang, Z. et al. Slot-die coating large-area formamidinium-cesium perovskite film for efficient and stable parallel solar module. *Sci. Adv.* **7**, eabg3749 (2021).
- Huang, X. et al. Solvent racing crystallization: low-solvation dispersion cosolvents for high-quality halide perovskites in photovoltaics. *Joule* **7**, 1556–1573 (2023).
- Siliavka, E. et al. Additive engineering of sequentially evaporated FAPbI₃ solar cells. *Adv. Energy Mater.* **16**, 2500963 (2025).
- Zhang, Z. et al. Efficient thermally evaporated γ -CsPbI₃ perovskite solar cells. *Adv. Energy Mater.* **11**, 2100299 (2021).

12. Lee, K. et al. Vacuum-processed propylene urea additive: a novel approach for controlling the growth of $\text{CH}_3\text{NH}_3\text{PbI}_3$ crystals in all vacuum-processed perovskite solar cells. *ACS Appl. Mater. Interfaces* **16**, 21915–21923 (2024).
13. Li, H. et al. Sequential vacuum-evaporated perovskite solar cells with more than 24% efficiency. *Sci. Adv.* **8**, eabo7422 (2022).
14. Niu, T. et al. Phase-pure α -FAPbI₃ perovskite solar cells via activating lead–iodine frameworks. *Adv. Mater.* **36**, 2309171 (2024).
15. Zhang, Y. et al. Nonalloyed α -phase formamidinium lead triiodide solar cells through iodine intercalation. *Science* **387**, 284–290 (2025).
16. Li, S. et al. High-efficiency and thermally stable FACsPbI₃ perovskite photovoltaics. *Nature* **635**, 82–88 (2024).
17. Liu, S. et al. Buried interface molecular hybrid for inverted perovskite solar cells. *Nature* **632**, 536–542 (2024).
18. Zai, H. et al. Wafer-scale monolayer MoS_2 film integration for stable, efficient perovskite solar cells. *Science* **387**, 186–192 (2025).
19. Xu, Y. et al. Boosted charge extraction of SnO_2 nanorod arrays via nanostructural and surface chemical engineering for efficient and stable perovskite solar cells. *Appl. Surf. Sci.* **607**, 154986 (2023).
20. Yao, Q. et al. Weakening solvent-solute interactions for high-efficiency screen-printed perovskite solar cells. *Angew. Chem. Int. Ed.* **64**, e202501350 (2025).
21. Sun, Y. et al. N-type surface design for p-type CZTSSe thin film to attain high efficiency. *Adv. Mater.* **33**, 2104330 (2021).
22. Tian, C. et al. Air-processed efficient perovskite solar cells with full lifecycle management. *Adv. Mater.* **37**, 2411982 (2025).
23. Zheng, Y. et al. Towards 26% efficiency in inverted perovskite solar cells via interfacial flipped band bending and suppressed deep-level traps. *Energy Environ. Sci.* **17**, 1153–1162 (2024).
24. Hamwi, S., Meyer, J., Winkler, T., Riedl, T. & Kowalsky, W. p-Type doping efficiency of MoO_3 in organic hole transport materials. *Appl. Phys. Lett.* **94**, 253307 (2009).
25. Khenkin, M. V. et al. Consensus statement for stability assessment and reporting for perovskite photovoltaics based on ISOS procedures. *Nat. Energy* **5**, 35–49 (2020).
26. Perdew, J. P., Burke, K. & Ernzerhof, M. Generalized gradient approximation made simple. *Phys. Rev. Lett.* **77**, 3865–3868 (1996).
27. Blöchl, P. E. Projector augmented-wave method. *Phys. Rev. B* **50**, 17953–17979 (1994).
28. Weller, M. T., Weber, O. J., Frost, J. M. & Walsh, A. Cubic perovskite structure of black formamidinium lead iodide, α -[$\text{HC}(\text{NH}_2)_2$]PbI₃, at 298 K. *J. Phys. Chem. Lett.* **6**, 3209–3212 (2015).
29. Lee, K., Murray, É.D., Kong, L., Lundqvist, B. I. & Langreth, D. C. Higher-accuracy van der Waals density functional. *Phys. Rev. B* **82**, 081101 (2010).
30. Lu, T. & Chen, F. Multiwfn: a multifunctional wavefunction analyzer. *J. Comput. Chem.* **33**, 580–592 (2012).
31. Ng, T.-W., Chan, C.-Y., Lo, M.-F., Guan, Z. Q. & Lee, C.-S. Formation chemistry of perovskites with mixed iodide/chloride content and the implications on charge transport properties. *J. Mater. Chem. A* **3**, 9081–9085 (2015).
32. Hu, H. et al. Vapour-based processing of hole-conductor-free $\text{CH}_3\text{NH}_3\text{PbI}_3$ perovskite/ C_{60} fullerene planar solar cells. *RSC Adv.* **4**, 28964–28967 (2014).
33. Scholz, L. et al. Hole-transport material variation in fully vacuum deposited perovskite solar cells. *APL Mater.* **2**, 081503 (2014).
34. Ke, W. et al. Efficient fully-vacuum-processed perovskite solar cells using copper phthalocyanine as hole selective layers. *J. Mater. Chem. A* **3**, 23888–23894 (2015).
35. Kim, B.-S., Kim, T.-M., Choi, M.-S., Shim, H.-S. & Kim, J.-J. Fully vacuum-processed perovskite solar cells with high open circuit voltage using MoO_3 /NPB as hole extraction layers. *Org. Electron.* **17**, 102–106 (2015).
36. Hsiao, S.-Y. et al. Efficient all-vacuum deposited perovskite solar cells by controlling reagent partial pressure in high vacuum. *Adv. Mater.* **28**, 7013–7019 (2016).
37. Momblona, C. et al. Efficient vacuum deposited p–i–n and n–i–p perovskite solar cells employing doped charge transport layers. *Energy Environ. Sci.* **9**, 3456–3463 (2016).
38. Kim, B.-S., Choi, M.-H., Choi, M.-S. & Kim, J.-J. Composition-controlled organometal halide perovskite via $\text{CH}_3\text{NH}_3\text{I}$ pressure in a vacuum co-deposition process. *J. Mater. Chem. A* **4**, 5663–5668 (2016).
39. Abzieher, T. et al. Efficient all-evaporated pin-perovskite solar cells: a promising approach toward industrial large-scale fabrication. *IEEE J. Photovoltaics* **9**, 1249–1257 (2019).
40. Longo, G. et al. Fully vacuum-processed wide band gap mixed-halide perovskite solar cells. *ACS Energy Lett.* **3**, 214–219 (2018).
41. Abzieher, T. et al. Electron-beam-evaporated nickel oxide hole transport layers for perovskite-based photovoltaics. *Adv. Energy Mater.* **9**, 1802995 (2019).
42. La-Placa, M.-G. et al. Vacuum-deposited 2D/3D perovskite heterojunctions. *ACS Energy Lett.* **4**, 2893–2901 (2019).
43. Palazon, F. et al. Room-temperature cubic phase crystallization and high stability of vacuum-deposited methylammonium lead triiodide thin films for high-efficiency solar cells. *Adv. Mater.* **31**, 1902692 (2019).
44. Babaei, A. et al. Preparation and characterization of mixed halide $\text{MAPbI}_{3-x}\text{Cl}_x$ perovskite thin films by three-source vacuum deposition. *Energy Technol.* **8**, 1900784 (2020).
45. Babaei, A. et al. Efficient vacuum deposited p–i–n perovskite solar cells by front contact optimization. *Front. Chem.* **7**, 936 (2020).
46. Pérez-del-Rey, D. et al. Molecular passivation of MoO_3 : band alignment and protection of charge transport layers in vacuum-deposited perovskite solar cells. *Chem. Mater.* **31**, 6945–6949 (2019).
47. Tavakoli, M. M., Yadav, P., Prochowicz, D. & Tavakoli, R. Efficient, hysteresis-free, and flexible inverted perovskite solar cells using all-vacuum processing. *Sol. RRL* **5**, 2000552 (2021).
48. Kim, B.-S., Gil-Escrig, L., Sessolo, M. & Bolink, H. J. Deposition kinetics and compositional control of vacuum-processed $\text{CH}_3\text{NH}_3\text{PbI}_3$ perovskite. *J. Phys. Chem. Lett.* **11**, 6852–6859 (2020).
49. Tavakoli, M. M. & Tavakoli, R. All-vacuum-processing for fabrication of efficient, large-scale, and flexible inverted perovskite solar cells. *Phys. Status Solidi R* **15**, 2000449 (2021).
50. Gil-Escrig, L. et al. Fully vacuum-processed perovskite solar cells on pyramidal microtextures. *Sol. RRL* **5**, 2000553 (2021).
51. Lei, T. et al. Flexible perovskite solar modules with functional layers fully vacuum deposited. *Sol. RRL* **4**, 2000292 (2020).
52. Paliwal, A. et al. Vacuum-deposited microcavity perovskite photovoltaic devices. *ACS Photonics* **8**, 2067–2073 (2021).
53. Abzieher, T. et al. From groundwork to efficient solar cells: on the importance of the substrate material in co-evaporated perovskite solar cells. *Adv. Funct. Mater.* **32**, 2201762 (2022).
54. Susic, I. et al. Intrinsic organic semiconductors as hole transport layers in p–i–n perovskite solar cells. *Sol. RRL* **6**, 2100882 (2022).
55. Choi, Y. et al. Toward all-vacuum-processable perovskite solar cells with high efficiency, stability, and scalability enabled by fluorinated spiro-ometad through thermal evaporation. *Sol. RRL* **5**, 2100415 (2021).
56. Kim, B.-S. et al. Simple approach for an electron extraction layer in an all-vacuum processed n–i–p perovskite solar cell. *Energy Adv.* **1**, 252–257 (2022).

57. Chen, C.-Y. et al. All-vacuum-deposited stoichiometrically balanced inorganic cesium lead halide perovskite solar cells with stabilized efficiency exceeding 11%. *Adv. Mater.* **29**, 1605290 (2017).
58. Lin, H.-Y. et al. Efficient cesium lead halide perovskite solar cells through alternative thousand-layer rapid deposition. *Adv. Funct. Mater.* **29**, 1905163 (2019).
59. Igual-Muñoz, A. M. et al. Room-temperature vacuum deposition of CsPbI₂Br perovskite films from multiple sources and mixed halide precursors. *Chem. Mater.* **32**, 8641–8652 (2020).
60. Pintor Monroy, M. I. et al. All-evaporated, all-inorganic CsPbI₃ perovskite-based devices for broad-band photodetector and solar cell applications. *ACS Appl. Electron. Mater.* **3**, 3023–3033 (2021).
61. Gil-Escrig, L. et al. Efficient vacuum-deposited perovskite solar cells with stable cubic FA_{1-x}MA_xPbI₃. *ACS Energy Lett.* **5**, 3053–3061 (2020).
62. Gil-Escrig, L. et al. Efficient wide-bandgap mixed-cation and mixed-halide perovskite solar cells by vacuum deposition. *ACS Energy Lett.* **6**, 827–836 (2021).
63. Gil-Escrig, L. et al. Vacuum deposited triple-cation mixed-halide perovskite solar cells. *Adv. Energy Mater.* **8**, 1703506 (2018).
64. Yuan, Q. et al. Thermally stable perovskite solar cells by all-vacuum deposition. *ACS Appl. Mater. Interfaces* **15**, 772–781 (2023).
65. Xu, Y.-Y. et al. Octahedral tilt enables efficient and stable fully vapor-deposited perovskite/silicon tandem cells. *Adv. Funct. Mater.* **34**, 2312037 (2024).
66. Murokinas, G. et al. Enhancing stability in all-vacuum-evaporated hybrid perovskite solar cells via a bipolar host as a hole-transporting layer. *ACS Appl. Mater. Interfaces* **17**, 5141–5152 (2025).
67. Feeney, T. et al. Understanding and exploiting interfacial interactions between phosphonic acid functional groups and co-evaporated perovskites. *Matter* **7**, 2066–2090 (2024).
68. Shih, C.-J. et al. Exploring buried interface in all-vapor-deposited perovskite photovoltaics. *Sol. Energy* **280**, 112872 (2024).
69. Diercks, A. et al. Sequential evaporation of inverted FAPbI₃ perovskite solar cells – impact of substrate on crystallization and film formation. *ACS Energy Lett.* **10**, 1165–1173 (2025).
70. Choi, Y. et al. Scalable all-vacuum-processed perovskite solar cells enabled by low energy-disorder hole-transport layer. *Adv. Energy Mater.* **15**, 2404797 (2025).
71. Peng, C. et al. High-performance thermally evaporated blue perovskite light-emitting diodes enabled by post-evaporation passivation. *Chem. Eng. J.* **499**, 155955 (2024).
72. Chen, B. et al. Self-assembled monolayers as hole transport layers for efficient thermally evaporated blue perovskite light-emitting diodes. *Chem. Eng. J.* **476**, 146476 (2023).
73. Peng, C. et al. The synergy of pre-frozen substrates and post-annealing for high-efficiency co-evaporated blue perovskite LEDs. *Chem. Eng. J.* **493**, 152579 (2024).
74. Sun, L. et al. Improved vacuum-evaporated blue perovskite light-emitting diodes with phenethylammonium chloride and guanidinium bromide synergistic post-processing modification. *Front. Optoelectron.* **18**, 5 (2025).
75. He, Z. et al. High-efficiency and emission-tunable inorganic blue perovskite light-emitting diodes based on vacuum deposition. *Small* **20**, 2305379 (2024).
76. Zhu, J. et al. All-thermally evaporated blue perovskite light-emitting diodes for active matrix displays. *Small Methods* **8**, 2300712 (2024).
77. Fu, Y. et al. Scalable all-evaporation fabrication of efficient light-emitting diodes with hybrid 2D–3D perovskite nanostructures. *Adv. Funct. Mater.* **30**, 2002913 (2020).
78. Jia, K. et al. Improved performance for thermally evaporated perovskite light-emitting devices via defect passivation and carrier regulation. *ACS Appl. Mater. Interfaces* **12**, 15928–15933 (2020).
79. Du, P. et al. Efficient and large-area all vacuum-deposited perovskite light-emitting diodes via spatial confinement. *Nat. Commun.* **12**, 4751 (2021).
80. Song, L. et al. Efficient thermally evaporated perovskite light-emitting devices via a bilateral interface engineering strategy. *J. Phys. Chem. Lett.* **12**, 6165–6173 (2021).
81. Hsieh, C.-A. et al. Vacuum-deposited inorganic perovskite light-emitting diodes with external quantum efficiency exceeding 10% via composition and crystallinity manipulation of emission layer under high vacuum. *Adv. Sci.* **10**, 2206076 (2023).
82. Peng, C. et al. The synergy of the buried interface surface energy and temperature for thermal evaporated perovskite light-emitting diodes. *ACS Appl. Mater. Interfaces* **15**, 15768–15774 (2023).
83. Meng, N. et al. Fully thermal-evaporated perovskite light-emitting diodes with brightness exceeding 240 000 nits. *Adv. Funct. Mater.* **35**, e10484 (2025).
84. Liu, N. et al. Fluorine-modified passivator for efficient vacuum-deposited pure-red perovskite light-emitting diodes. *Light Sci. Appl.* **14**, 118 (2025).
85. He, S. et al. Efficient thermally evaporated near-infrared perovskite light-emitting diodes via phase regulation. *Nano Micro Lett.* **17**, 270 (2025).
86. Dänekamp, B. et al. Efficient photo- and electroluminescence by trap states passivation in vacuum-deposited hybrid perovskite thin films. *ACS Appl. Mater. Interfaces* **10**, 36187–36193 (2018).

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Author contributions

Y. Chen and Q.G. conceived the idea and directed the overall project. W.H., Y. Chen, Q.G. and Y. Xia supervised the work. Y. Xu fabricated all the devices and conducted the characterization. T.P. helped with the device fabrication and material characterization. Y.L., X.K. and N.M. performed the PL and EL imaging characterization. J.L., Y. Cui and Z.W. performed the atomic Kelvin probe force microscopy imaging characterization. K.X. and X.S. performed the DFT and molecular dynamics calculations. X.M., L.C., Z.H. and N.W. participated in data analysis. Y. Xia, Q.G. and Y. Chen wrote the paper. All authors read and commented on the paper.

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Competing interests

The authors declare no competing interests.

Additional information

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Solar Cells Reporting Summary

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► Experimental design

Please check the following details are reported in the manuscript, and provide a brief description or explanation where applicable.

1. Dimensions

Area of the tested solar cells

☒ Yes
☐ No

The area of tested solar cells is 0.066 cm².

Explain why this information is not reported/not relevant.

Method used to determine the device area

☒ Yes
☐ No

Metal aperture mask.

Explain why this information is not reported/not relevant.

2. Current-voltage characterization

Current density-voltage (J-V) plots in both forward and backward direction

☐ Yes
☒ No

Not relevant to this work.

Voltage scan conditions

☒ Yes
☐ No

Section "Device measurement".

Explain why this information is not reported/not relevant.

Test environment

☒ Yes
☐ No

Section "Device measurement".

Explain why this information is not reported/not relevant.

Protocol for preconditioning of the device before its characterization

☐ Yes
☒ No

Provide a description of the protocol.

No preconditioning is required for our cell characterization.

Stability of the J-V characteristic

☒ Yes
☐ No

Section "Stability characterization", Fig. 4j.

Explain why this information is not reported/not relevant.

3. Hysteresis or any other unusual behaviour

Description of the unusual behaviour observed during the characterization

☐ Yes
☒ No

Provide a description of hysteresis or any other unusual behaviour observed during the characterization.

No unusual behavior was observed.

Related experimental data

☐ Yes
☒ No

Provide a description of the related experimental data.

No unusual behavior was observed.

4. Efficiency

External quantum efficiency (EQE) or incident photons to current efficiency (IPCE)

☒ Yes
☐ No

We provide EQE measurement in Supplementary Fig. 55.

Explain why this information is not reported/not relevant.

A comparison between the integrated response under the standard reference spectrum and the response measure under the simulator

☒ Yes
☐ No

The integrated J_{sc} values are consistent with those from J-V measurements.

Explain why this information is not reported/not relevant.

For tandem solar cells, the bias illumination and bias voltage used for each subcell	☐ Yes ☒ No	Provide a description of the measurement conditions. We didn't fabricate tandem solar cell.
5. Calibration		
Light source and reference cell or sensor used for the characterization	☒ Yes ☐ No	See details in Method. Explain why this information is not reported/not relevant.
Confirmation that the reference cell was calibrated and certified	☒ Yes ☐ No	Our solar simulator was calibrated by Si reference cell. Explain why this information is not reported/not relevant.
Calculation of spectral mismatch between the reference cell and the devices under test	☐ Yes ☒ No	Provide a value of the spectral mismatch and/or a description of how it has been taken into account in the measurements. No spectral mismatch calculation was performed.
6. Mask/aperture		
Size of the mask/aperture used during testing	☒ Yes ☐ No	An optical aperture mask (0.066 cm²) was used. Explain why this information is not reported/not relevant.
Variation of the measured short-circuit current density with the mask/aperture area	☐ Yes ☒ No	Report the difference in the short-circuit current density values measured with the mask and aperture area. No significant variations was observed.
7. Performance certification		
Identity of the independent certification laboratory that confirmed the photovoltaic performance	☒ Yes ☐ No	Certified PCE of 24.93% for the 0.0659 cm² PSC was achieved from an independently accredited testing centre (Tianjin Institute of Metrological Supervision and Testing, China) (Supplementary Fig. 56). Explain why this information is not reported/not relevant.
A copy of any certificate(s)	☒ Yes ☐ No	We have provided the certified efficiency in Supplementary Fig. 56. Explain why this information is not reported/not relevant.
8. Statistics		
Number of solar cells tested	☒ Yes ☐ No	Supplementary Figs. 52, 57 and 59. Explain why this information is not reported/not relevant.
Statistical analysis of the device performance	☒ Yes ☐ No	Supplementary Figs. 52, 57 and 59. Explain why this information is not reported/not relevant.
9. Long-term stability analysis		
Type of analysis, bias conditions and environmental conditions	☒ Yes ☐ No	We provided stability data in Fig.4j. Explain why this information is not reported/not relevant.